Tarmoq qurilmalarini diagnostika qilish

Basic Network Device Debugging

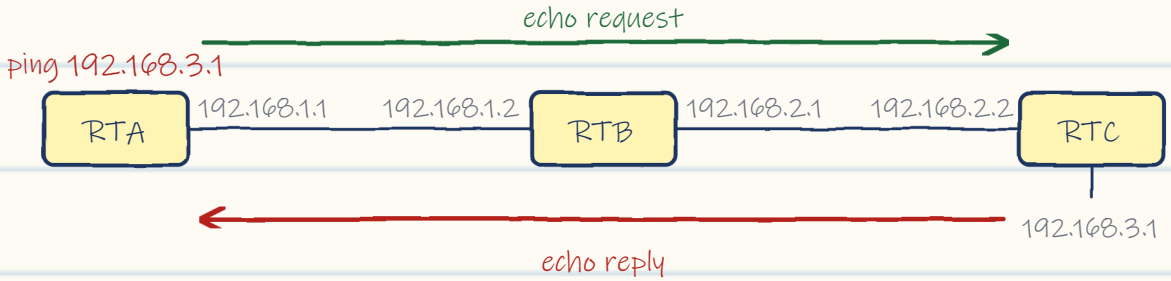
Asosiy atamalar

- ping — ICMP echo bilan ulanish borligini tekshiradi
- tracert — paket o'tgan routerlar yo'lini ko'rsatadi
- TTL — paket umri, har bir routerda 1 ga kamayadi
- debugging — tizim ishini batafsil kuzatish rejimi

1. ping — ulanishni tekshirish

ICMP ustida ishlaydi: manba echo request, manzil echo reply.

- javob keldi — manzilning IP qatlami ishlayapti
- javob yo'q — destination unreachable yoki timeout



1-rasm. RTA → RTC: so'rov ketadi, javob qaytadi



ICMP — IP ichida, protocol = 1

2. ping natijasini o‘qish

```
[RTA]ping 192.168.3.1
56 bytes from 192.168.3.1:
icmp_seq=0 ttl=64 time=1.8 ms
---
5 packets transmitted,
5 packets received,
0.0% packet loss
min/avg/max = 0.9/1.4/1.8 ms
```

icmp_seq — tartib raqami

ttl — qolgan hop zaxirasi

time — borib-kelish vaqti

packet loss — yo‘qotish %

sukut bo‘yicha 5 ta so‘rov

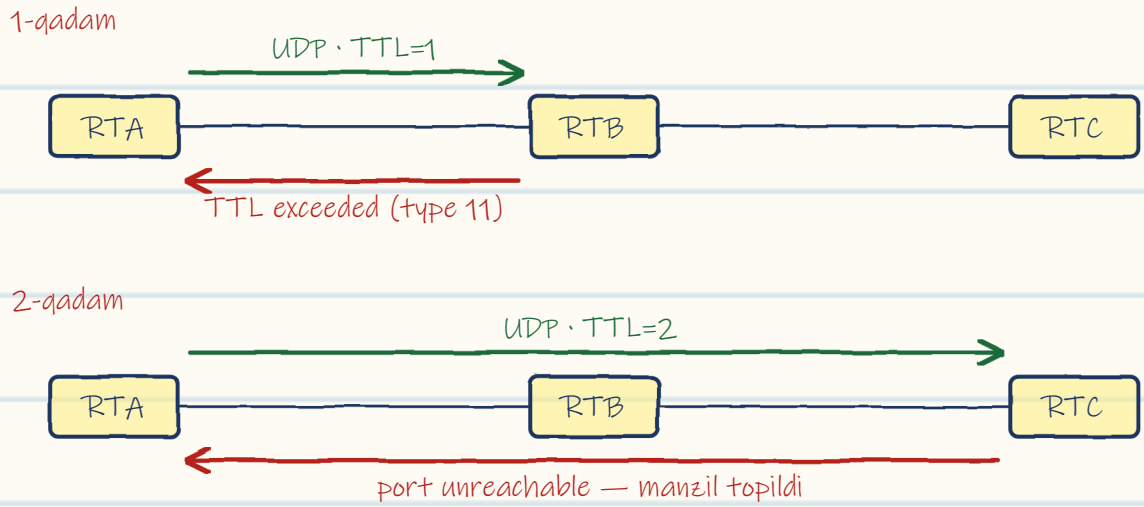
2-rasm. Har satr — bitta javob, oxirida statistika

3. ping kalitlari

-c	so‘rov soni · 5	-s	uzunlik · 56 bayt
-t	timeout · 2000 ms	-h	TTL · 255
-m	interval · 200 ms	-a	manba IP

-f — bo‘lakka bo‘lmaslik (MTU sinovi) · -q — faqat statistika

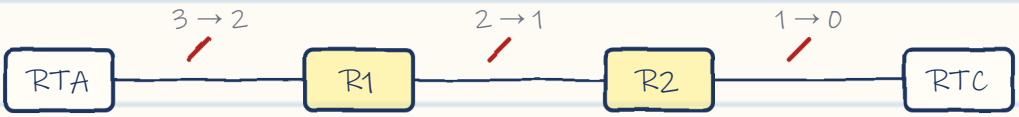
4. tracert — yo‘lni ko‘rish



5. tracert mexanizmi

- TTL 1 dan boshlanadi va har so‘rovda 1 ga oshadi
- har router TTL ni 1 ga kamaytiradi; 0 bo‘lsa — TTL exceeded
- manzilga yetsa — port unreachable, izlash to‘xtaydi
- shu yo‘l bilan har hopning IP manzili aniqlanadi

TTL = 0 bo‘lgan joy o‘zini bildiradi



har hop TTL ni 1 ga kamaytiradi

```
[RTA]tracert 192.168.3.1
30 hops at most, 52 bytes
1 192.168.1.2 0.69 0.50 0.49 ms
2 192.168.3.1 1.00 0.90 0.90 ms
* — javob kelmadi
```

1-hop = RTB

2-hop = manzil

har hopga 3 ta sinov

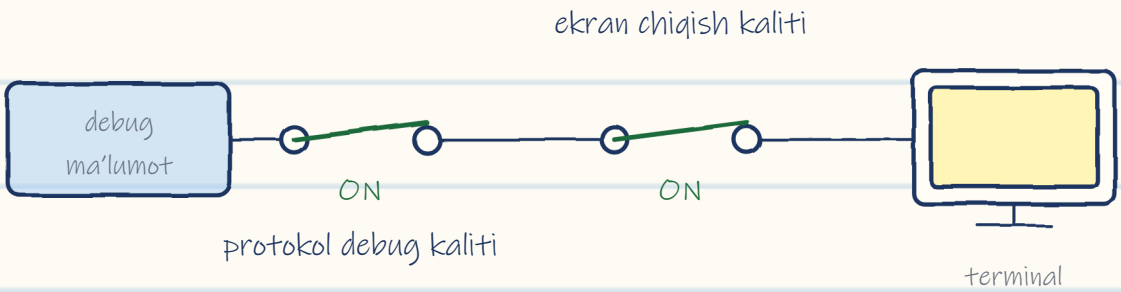
uchta vaqt — uch urinish

3-rasm. Yo‘ldagi har tugun bir satr bo‘lib chiqadi

-f	boshlang‘ich TTL · 1	-m	maks TTL · 30
-q	sinov soni · 3	-p	UDP port · 33434
-w	timeout · 5000 ms	-a	manba IP

6. Tizim diagnostikasi (debugging)

Ko‘p protokol va modullar o‘z diagnostika ma’lumotini beradi. Chiqishni ikki kalit boshqaradi.



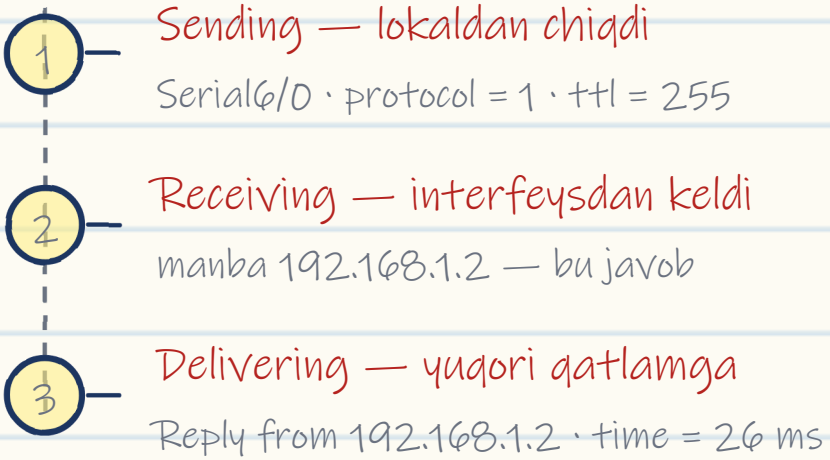
ikkisi ham ON → ekranda ko'rinadi

protokol kaliti	ekran kaliti	natija
OFF	ON	chiqmaydi
ON	OFF	chiqmaydi
ON	ON	chiqadi

7. Buyruqlar

terminal monitor	konsol monitoringi (sukutda yoniq)
terminal debugging	shu terminalga chiqarish
debugging modul	modul kaliti, masalan debugging ip packet
display debugging	yoqilganlar ro'yxati

8. debugging ip packet natijasi



4-rasm. Bitta ping uchta qayd beradi: chiqish · kelish · yetkazish

Xulosa

- ping = ICMP echo request + echo reply
- ping standarti: 5 so'rov · 56 bayt · timeout 2000 ms
- tracert = TTL ni oshirib borish; TTL exceeded → har hop bilinadi
- tracert standarti: maks TTL 30 · 3 sinov · UDP 33434
- debug chiqishi: protokol kaliti va ekran kaliti — ikkisi ham kerak

Tekshirish

1. ping qaysi protokolda? — ICMP (protocol 1)
2. tracert nimadan foydalanadi? — TTL tugashi
3. tracert maks TTL sukutda? — 30
4. Oraliq router nima qaytaradi? — TTL exceeded, type 11
5. debug ko'rinishi uchun nima kerak? — ikki kalit ON